QUICKSTART GUIDE

This section provides a brief example of ADS1113/4/5 communications. Refer to subsequent sections of this data sheet for more detailed explanations. Hardware for this design includes: one ADS1113/4/5 configured with an I²C address of 1001000; a microcontroller with an I²C interface (TI recommends the MSP430F2002); discrete components such as resistors, capacitors, and serial connectors; and a 2V to 5V power supply. Figure 23 shows the basic hardware configuration.

The ADS1113/4/5 communicate with the master (microcontroller) through an I²C interface. The master provides a clock signal on the SCL pin and data are transferred via the SDA pin. The ADS1113/4/5 never drive the SCL pin. For information on programming and debugging the microcontroller being used, refer to the device-specific product data sheet.

The first byte sent by the master should be the ADS1113/4/5 address followed by a bit that instructs the ADS1113/4/5 to listen for a subsequent byte. The second byte is the register pointer. Refer to Table 9 for a register map. The third and fourth bytes sent from the master are written to the register indicated in the second byte. Refer to Figure 30 and Figure 31 for read and write operation timing diagrams, respectively. All read and write transactions with the ADS1113/4/5 must be preceded by a start condition and followed by a stop condition.

For example, to write to the configuration register to set the ADS1113/4/5 to continuous conversion mode and then read the conversion result, send the following bytes in this order:

Write to Config register:

First byte: 0b10010000 (first 7-bit I²C address followed by a low read/write bit)

Second byte: 0b00000001 (points to Config register)

Third byte: 0b10000100 (MSB of the Config register to be written)

Fourth byte: 0b10000011 (LSB of the Config register to be written)

Write to Pointer register:

First byte: 0b10010000 (first 7-bit I²C address followed by a low read/write bit)

Second byte: 0b00000000 (points to Conversion register)

Read Conversion register:

First byte: 0b10010001 (first 7-bit I²C address followed by a high read/write bit)

Second byte: the ADS1113/4/5 response with the MSB of the Conversion register

Third byte: the ADS1113/4/5 response with the LSB of the Conversion register

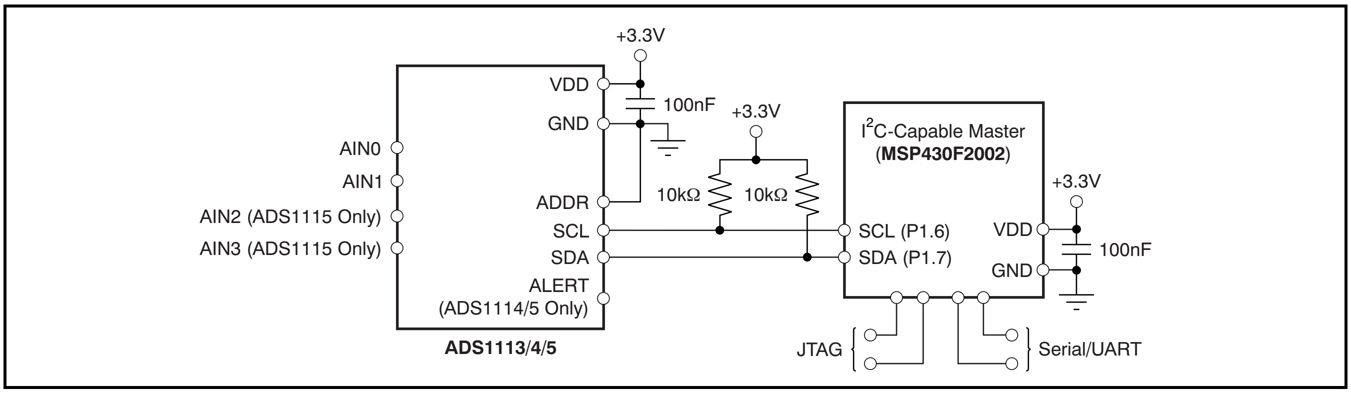

Figure 23. Basic Hardware Configuration

MULTIPLEXER

The ADS1115 contains an input multiplexer, as shown in Figure 24 . Either four single-ended or two differential signals can be measured. Additionally, AIN0 and AIN1 may be measured differentially to AIN3. The multiplexer is configured by three bits in the Config register. When single-ended signals are measured, the negative input of the ADC is internally connected to GND by a switch within the multiplexer.

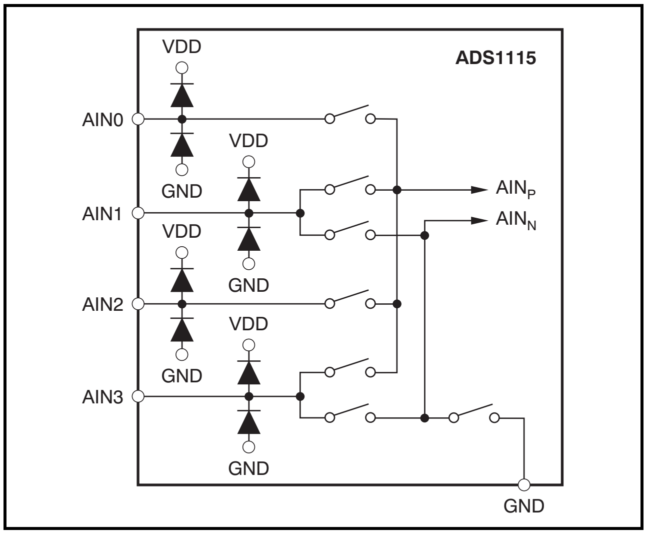

Figure 24. ADS1115 MUX

The ADS1113 and ADS1114 do not have a multiplexer. Either one differential or one single-ended signal may be measured with these devices. For single-ended measurements, connect the AIN1 pin to GND. Note that in subsequent sections of this data sheet, AIN_P refers to AIN0 and AIN_N refers to AIN1 for the ADS1113 and ADS1114.

When measuring single-ended inputs it is important to note that the negative range of the output codes are not used. These codes are for measuring negative differential signals such as $(AIN_P – AIN_N) < 0$. ESD diodes to VDD and GND protect the inputs on all three devices (ADS1113, ADS1114, and ADS1115). To prevent the ESD diodes from turning on, the absolute voltage on any input must stay within the following range:

GND – 0.3V < AINx < VDD + 0.3V

If it is possible that the voltages on the input pins may violate these conditions, external Schottky clamp diodes and/or series resistors may be required to limit the input current to safe values (see the Absolute Maximum Ratings table).

Also, overdriving one unused input on the ADS1115 may affect conversions taking place on other input pins. If overdrive on unused inputs is possible, again it is recommended to clamp the signal with external Schottky diodes.

ANALOG INPUTS

The ADS1113/4/5 use a switched-capacitor input stage where capacitors are continuously charged and then discharged to measure the voltage between AIN_P and AIN_N. The capacitors used are small, and to external circuitry the average loading appears resistive. This structure is shown in Figure 26 . The resistance is set by the capacitor values and the rate at which they are switched. Figure 25 shows the on/off setting of the switches illustrated in Figure 26 . During the sampling phase, S_1 switches are closed. This event charges C_{A1} to AIN_P, C_{A2} to AIN_N, and C_B to $(AIN_P – AIN_N)$. During the discharge phase, S_1 is first opened and then S_2 is closed. Both C_{A1} and C_{A2} then discharge to approximately 0.7V and C_B discharges to 0V. This charging draws a very small transient current from the source driving the ADS1113/4/5 analog inputs. The average value of this current can be used to calculate the effective impedance (R_{eff}) where $R_{eff} = V_{IN}/I_{AVERAGE}$.

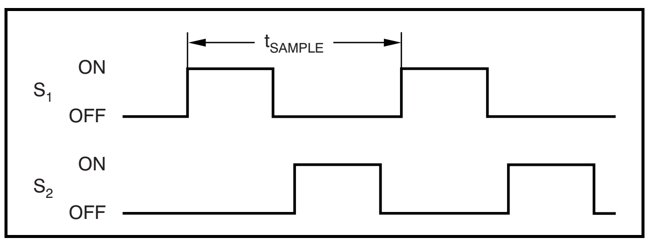

Figure 25. S_1 and S_2 Switch Timing for Figure 26